PARENT ROUNDTABLE SERIES

The Why Behind Supplements

How a pharmacist evaluates a foundational supplement for quality, evidence, and safety

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SETTING THE STAGE

A supplement is a tool, not a cure

- In the U.S., supplements are regulated as foods, not drugs
- They are not FDA-approved for safety or effectiveness before they are sold
- Labels may describe support for a body function, but may not claim to treat, cure, or prevent a disease

What that means for families

The burden of choosing well falls on you and your care team. A pharmacist's review of quality, evidence, and interactions closes that gap.

A PHARMACIST'S FRAMEWORK

Five questions I ask before recommending any supplement

1

What is it?

Form and dose

2

Why this child?

Need or deficiency

3

What is the evidence?

For this use

4

Is it quality-made?

Testing and COA

5

Is it safe?

With medicines

> If any answer is unclear, the right recommendation is to wait and find out.

QUALITY IS ON PAPER

Quality you can verify: the label is only the start

Manufacturing standards

Makers must follow current good manufacturing practice rules for identity, purity, strength, and composition.

Certificate of Analysis

A batch-level report confirming identity, potency, and limits for contaminants such as heavy metals and microbes.

Third-party seals

Independent programs such as USP Verified and NSF certification test products against their labels.

> In my product work, I reviewed batch Certificates of Analysis before products reached families.

Dietary Supplement Health and Education Act of 1994; 21 CFR Part 111 (current good manufacturing practice)

What the evidence supports for common foundational supplements

Supplement	What it does	Evidence note
Probiotics	Support the balance of gut microbes	Effects are strain-specific; strongest pediatric evidence is for preventing antibiotic-associated diarrhea
Omega-3 fats	Structural part of cell membranes, especially in the brain and retina	Food sources first; dose depends on purpose
Vitamin D	Bone health and immune function	Test the level, then supplement to target
Magnesium	Cofactor in more than 300 enzyme systems, including muscle and nerve function	Forms differ in absorption and gut tolerance
Zinc	Immune function and wound healing	Long-term excess can cause copper deficiency

NIH ODS fact sheets; Guo Q, et al. Cochrane Database Syst Rev. 2019. PMID 31039287

INTERACTIONS

Check interactions before adding anything new

Supplement	Interacts with	What to do
Magnesium, zinc	Tetracycline and quinolone antibiotics	Separate doses by several hours; ask the pharmacist for timing
Omega-3 (high dose)	Blood thinners	Ask the prescriber before starting
Vitamin D	Thiazide diuretics	Calcium can rise; monitor as advised
Probiotics	Weakened immune system, central lines	Use only with the care team's approval

> Bring every supplement bottle to medication reviews. Interactions are only caught when the list is complete.

MORE IS NOT BETTER

Every nutrient has an upper limit, and children's limits are lower

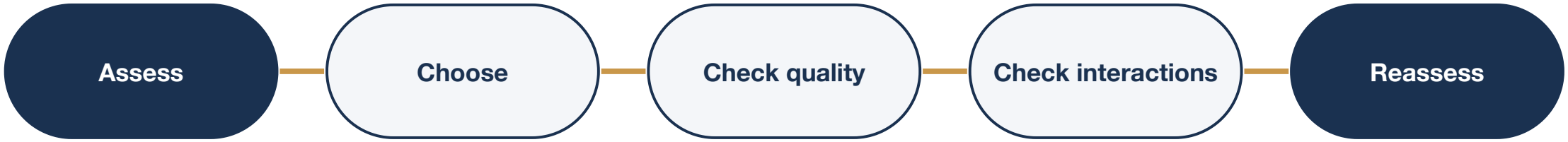
- Upper limits are set by age; a child's limit can be a fraction of an adult's
- Excess vitamin D can raise calcium to unsafe levels
- Excess zinc can cause copper deficiency
- Stacking products can double a dose without anyone noticing

Pharmacist tip

Add up the same nutrient across every product: the multivitamin, the gummy, and the fortified drink all count.

BRINGING IT TOGETHER

Assess, choose carefully, then reassess



- Start from a real need, confirmed by history or testing
- Change one thing at a time so you can tell what helped
- Reassess on a schedule, and stop what is not working

REFERENCES

Sources

1. Dietary Supplement Health and Education Act of 1994; 21 CFR Part 111 (current good manufacturing practice)
2. NIH Office of Dietary Supplements fact sheets for health professionals (ods.od.nih.gov)
3. Guo Q, et al. Cochrane Database Syst Rev. 2019;4:CD004827. Probiotics for the prevention of pediatric antibiotic-associated diarrhea. PMID 31039287
4. Holick MF, et al. J Clin Endocrinol Metab. 2011;96(7):1911-30. PMID 21646368